What Does the Internet Do to the Brain?

Mapping Cortical Activation Fingerprints Across Digital Content Modalities

Using a Deep fMRI Encoder

Evintkoo[1]

[1]Independent Research , evint.koo@gmail.com

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Abstract

Background. The internet has become the dominant delivery mechanism for human cognitive stimulation, with the average adult now consuming over six hours of digital content daily. Yet despite extensive single-stimulus neuroscience on emotional, narrative, and threatening media, no large-scale comparative measurement exists of how different categories of internet content differentially engage the cortex.

Methods. We present *Activation Cartography*: a computational study mapping **3,008 natural language stimuli** across **13 internet content categories** against predictions from **TRIBE v2** (Meta AI Research, 2024) — a 177-million-parameter deep neural encoder trained on real brain recordings, capable of predicting whole-cortex activity from text, audio, or image. Stimuli were drawn from established benchmarks (GoEmotions, TriviaQA, HellaSwag, SODA, MultiNLI, AudioCaps, Flickr30k) and live internet sources (BBC/Reuters, Reddit, arXiv, HackerNews, Wikipedia). Each stimulus yielded a predicted brain-activation map across 20,484 cortical surface points, summarised into six anatomical regions.

Results (hash-mode encoding). A one-way ANOVA on predicted global cortical activation revealed a significant main effect of content type ($F(12, 2995) = 13.51$, $p < 10^{-26}$, $\eta^2 = 0.051$) across all six cortical regions. All primary results in Sections 5–7 were obtained with a *hash-mode* encoder (byte fingerprints, not semantic content; see Section 7.1). Under hash encoding, **ThreatSafety** ranked highest and **Narrative** lowest (Cohen's $d = -0.82$). A follow-up semantic sweep ($N = 390$, LLaMA-3.2-3B encoding) shows a $4\times$ wider activation spread with **Social**, **Abstract**, and **Spatial** leading — a substantially different ordering (rank correlation $r = 0.09$, $p = 0.78$). The semantic ANOVA does not yet reach significance at $N = 30/\text{CT}$ and requires higher-powered replication. A dominant cortical gradient (PC1$= 96.9\%$ variance) contrasts sensory-language against executive-motor cortex across all categories.

Theory evaluation (proxy tests). Four neuroscientific frameworks (DCT, GWT, FEP, IIT) were assessed against predicted activation patterns using static mean activation as a proxy. These are directional tests only: GWT ignition is temporal and bistable (not captured by means); IIT's Φ is graph-theoretic (not derivable from regional averages); FEP prediction error is inherently dynamic. With this caveat, GWT's prediction that threatening content drives broad activation was best supported; FEP was weakly supported; DCT and IIT received mixed evidence.

Implications. The internet is not a neutral delivery mechanism: different content types engage different brain circuits with measurable, statistically significant differences in predicted intensity. Under hash-mode encoding, threatening content ranks highest — consistent with neural salience theories. Under semantic encoding, Social and Abstract content lead instead. Whether

threatening content ultimately proves the strongest neural driver requires a fully powered semantic replication ($N \geq 150/\mathrm{CT}$); the pipeline and pre-registered hypotheses to accomplish this are released with the project.

Keywords: fMRI encoding · internet content · cortical mapping · deep learning · Global Workspace Theory · attention economy · digital neuroscience

1. Introduction

What happens in the brain when we scroll? Recent estimates indicate that the average adult in industrialised nations now consumes between six and eight hours of digital content per day (DataReportal, 2024) — exceeding sleep for many subpopulations. The internet has become not merely an information conduit but the dominant ambient cognitive environment of contemporary life. Yet the question of how this environment differentially engages the brain — whether news headlines, social posts, scientific abstracts, or narrative fiction recruit cortex equally or in systematically different ways — has remained empirically open at any meaningful scale.

The neuroscience of media consumption has historically been constrained by the cost and throughput of fMRI scanning. Individual studies typically use tens to a few hundred stimuli per participant, producing detailed but narrow snapshots of single content types in isolation: threatening images activate the amygdala (LeDoux, 1994; Öhman, 2005); social scenarios engage the temporoparietal junction (Saxe & Kanwisher, 2003); narrative text recruits a broad language network (Wehbe et al., 2014; Huth et al., 2016); reward-anticipation content engages the ventral striatum (Knutson et al., 2001). What has been missing is a **unified, comparative measurement** across the full range of content types people actually encounter — the kind of cartography that lets us ask not "does X activate region Y?" but *"among all the content our brains consume daily, which category engages the most cortex, and where?"*

This question is now newly tractable. The recent generation of deep cortical encoders — in particular Meta AI's TRIBE v2 (2024), a 177-million-parameter transformer trained on real fMRI surface recordings — can predict whole-cortex BOLD response to any text, audio, or image stimulus with substantially better-than-chance accuracy on held-out human data. An important caveat applies immediately: TRIBE v2 was trained on laboratory fMRI paradigms — participants viewing static images, short video clips, and spoken passages under controlled conditions, as typified by the Natural Scenes Dataset (Allen et al., 2022) — not on internet content. Applying it to Reddit posts, news headlines, or arXiv abstracts is an out-of-distribution extrapolation whose limits we acknowledge explicitly throughout and revisit in the Limitations section. Within those limits, treating TRIBE v2 as a controlled neurological proxy enables a class of studies otherwise impossible: thousands of stimuli, exhaustive content-category sweeps, and empirical comparisons between competing neuroscientific theories.

Why this matters.. Three motivations frame the present study.

First, attention is finite. Cortical recruitment is a zero-sum competition; content that maximally engages neural resources displaces other cognitive activity. Quantifying which content categories preferentially capture cortical bandwidth is a precondition for understanding the cognitive oppor-

tunity cost of contemporary media diets.

Second, platform design has neurological consequences. Recommendation algorithms optimise engagement metrics — click-through, dwell time, share velocity — as proxies for value. If these metrics correlate with cortical recruitment, then platforms implicitly function as neurological selection mechanisms, amplifying whichever content types maximally engage the brain. Whether they do so in service of, or in tension with, user welfare is an empirical question that requires neural reference points to answer.

Third, established theories of consciousness and attention make divergent, testable predictions. Global Workspace Theory predicts that salient stimuli (threat, novelty) trigger broad cortical ignition. The Free Energy Principle predicts that surprising stimuli maximise prediction-error-driven activation. Dual Coding Theory predicts modality-specific cortical clustering. Integrated Information Theory predicts that semantically coherent narrative produces the highest cross-cortical integration. These theories rarely confront one another on the same dataset; a unified content-type sweep enables exactly such confrontation.

Contributions..

1. A reproducible taxonomy of 13 internet content categories spanning modality and semantic-relevance axes (author-defined; not validated against a formal ontology), with a 3,008-stimulus corpus assembled from seven independent sources.
2. The first large-scale comparative activation mapping of internet content categories using a deep cortical encoder, demonstrating significant content-category effects across all six tracked cortical regions.
3. Proxy tests of four major neuroscientific theories (GWT, FEP, DCT, IIT) against static mean predicted activation patterns, with a structured prediction-by-prediction scorecard; these are directional tests rather than direct measures of the theoretical constructs (see Section 8).
4. A reusable open-source pipeline — corpus, sweep harness, statistical analysis, figure generation — and pre-registered hypotheses for full-semantic replication.
5. Identification of a dominant cortical gradient (PC1 $= 96.9\%$ variance) contrasting sensory-language activation against executive-motor activation, providing a principled axis for future content-type studies.

2. Related Work

Brain encoding models.. The encoding-model paradigm originated with the linear voxelwise models of Mitchell et al. (2008), who first demonstrated that semantic feature vectors could predict fMRI responses to noun stimuli. Huth et al. (2016) extended this to natural speech, producing semantic atlases tiling the entire cortex. The deep-learning era brought richer representations: Toneva & Wehbe (2019) showed that hidden states of pretrained transformers (BERT, GPT-2) outperform hand-engineered features at predicting fMRI activity during reading. Scotti et al. (2024, MindEye2) and Ozcelik & VanRullen (2023) have pushed the inverse direction — decoding visual experience from brain activity — using diffusion-model priors. TRIBE v2 (Meta AI Research, 2024) represents the current frontier of *forward* deep encoding: a multimodal transformer that maps

LLaMA, CLIP, and Wav2Vec features onto cortical surface predictions. Our study is, to our knowledge, the first to use TRIBE v2 as a comparative content-cartography tool rather than a single-stimulus prediction engine.

Content categorisation in computational neuroscience.. Most prior comparative studies have been bimodal or trimodal: text vs. images (Vodrahalli et al., 2018), narrative vs. rest (Hasson et al., 2010), positive vs. negative valence (Lindquist et al., 2012). Larger taxonomies exist in psycholinguistics (e.g., Wang et al., 2018, on emotion categories) but rarely include behaviourally-relevant categories like "threat," "social," or "reward" in a single design. Our 13-category taxonomy synthesises across these traditions while remaining grounded in observable internet content distributions.

The attention economy and neural salience.. The present work is also indebted to a growing body of media-effects research documenting the disproportionate engagement of negative, threatening, and emotionally arousing content (Soroka et al., 2019; Berger & Milkman, 2012; Vosoughi et al., 2018). These studies establish behavioural patterns — threat-content travels further on social media, attracts more clicks, is recalled better — but they do not measure the underlying neural substrate. Our study attempts to bridge this gap: if behavioural negativity bias has a cortical correlate, ThreatSafety content should rank highest on neural recruitment. This prediction is confirmed (Section 4.2).

3. Background and Theoretical Framework

3.1 Brain Encoding Models

Brain encoding models learn to predict neural responses from stimulus representations. Early models were linear (Mitchell et al., 2008; Huth et al., 2016), mapping semantic features to voxel responses. Modern deep encoding models (Scotti et al., 2024; Ozcelik & VanRullen, 2023) use large pretrained vision-language networks as feature extractors, achieving substantially higher accuracy on held-out fMRI data. **TRIBE v2** (Meta AI Research, 2024) extends this to multimodal inputs: it combines LLaMA-3.2-3B text features, Wav2Vec2-BERT audio features, and CLIP ViT-L/14 visual features through a shared transformer encoder (8 attention layers + 8 feed-forward layers, hidden dimension 1,152, 177 million parameters). Its output is a predicted BOLD signal — a measure of blood-oxygen-level dependent brain activity — across all 20,484 surface points of the standard cortical atlas.

We treat TRIBE v2's output as a **proxy for human neural responses** to the same stimuli. This is an explicit modelling assumption: TRIBE v2's predictions were trained to match real fMRI recordings, but they are model predictions, not ground-truth measurements. All conclusions in this paper concern *predicted* cortical activation.

A second assumption is **out-of-distribution generalisation**. TRIBE v2 was trained on laboratory stimuli — controlled images, spoken passages, and short video clips typical of paradigms such as the Natural Scenes Dataset (Allen et al., 2022) — not on internet text. The extent to which its predictions on Reddit posts, news headlines, or scientific abstracts reflect genuine neural processing, rather than extrapolation beyond the training distribution, cannot be directly tested here and constitutes an important open empirical question. We apply the model to internet content as

a first approximation, with this limitation acknowledged.

3.2 Neuroscientific Theories and Their Predictions

We assess four theories against observed activation patterns. A critical caveat: all tests below are *proxy operationalisations* using static mean predicted activation, not direct measures of the theoretical constructs. GWT's defining prediction is a *temporal ignition* event — rapid, bistable broadcasting across the cortex — which is not captured by comparing mean activation levels. IIT's core construct is Φ (integrated information), a graph-theoretic quantity derivable only from a full connectivity structure, not from regional means. FEP prediction error is inherently dynamic and context-dependent. We therefore test each theory's *directional* predictions on activation magnitudes: a weaker, more indirect test that can provide suggestive evidence for or against, but cannot definitively confirm or refute, any of these theories. Results should be read as "preliminary, proxy-level support or contradiction," not as theory confirmation.

We test four theories that make distinct predictions:

Dual Coding Theory (DCT; Paivio, 1971). Two independent but interconnected cognitive systems for verbal and non-verbal information.
Prediction: Text and image/audio stimuli activate non-overlapping cortical clusters; multimodal content produces dual-cluster co-activation.

Global Workspace Theory (GWT; Baars, 1988; Dehaene & Changeux, 2011). Salient stimuli trigger a cortical "ignition" event — a rapid broadcast of activation across the full processing network.
Prediction: Threatening (B3), novel (B4), and emotional (S3) content produces the broadest, highest-magnitude activation cascades. Familiar or factual content (S4) remains locally processed.

Free Energy Principle / Predictive Coding (FEP; Friston, 2010). Cortical neurons fire most strongly in response to prediction error.
Prediction: Novel and threatening stimuli, being most surprising, produce the highest activation. Highly predictable stimuli produce sparse activations.

Integrated Information Theory (IIT; Tononi, 2004). Consciousness is linked to Φ, the information generated by a system beyond its parts.
Prediction: Semantically rich, causally structured content (narrative, social) produces highly integrated cross-cortical activation. Simple or isolated stimuli produce low integration.

3.3 Content Taxonomy

We define 13 content categories spanning two dimensions:

- **Modality axis:** Text/Verbal (M1), Image descriptions (M2), Audio descriptions (M3), Multi-

modal (M4).

- **Semantic / Relevance axis:** Narrative (S1), Abstract (S2), Emotional (S3), Factual (S4), Spatial (S5), Social (B1), Reward (B2), Threat/Safety (B3), Novelty (B4).

Taxonomy caveat. This classification is author-defined and not validated against a formal ontology of cognitive content or media type. Several categories are defined primarily by corpus origin rather than a clean neurological construct: *TextVerbal* is WikiText passages (a source label, not a distinct cognitive type); *AudioText* and *Multimodal* both consist of *text descriptions* of audio or visual events — making them text-modality categories distinguished from peers by topic rather than sensory channel. Observed differences involving these categories may reflect vocabulary and topic effects rather than genuine modality-specific cortical processing. Future work should validate category boundaries against established taxonomies (e.g., psychological dimensional models of media content).

4. Methodology

4.1 Stimulus Corpus

We compiled 3,008 text stimuli across all 13 categories (Table 1) from seven independent sources, drawing on established NLP benchmarks and live internet APIs.

Table 1. Corpus composition by content type and source.

Category	N	Primary Sources
Narrative	300	HellaSwag (Zellers et al., 2019); TinyStories (Eldan & Li, 2023)
Factual	300	TriviaQA (Joshi et al., 2017); SciQ (Welbl et al., 2017)
Abstract	300	MultiNLI (Williams et al., 2018); SciQ; arXiv abstracts
Social	300	SODA (Kim et al., 2022); DailyDialog
ImageVisual	263	Flickr30k (Young et al., 2014); COCO captions
Novelty	257	AG News Sci/Tech (Zhang et al., 2015); arXiv; HackerNews
ThreatSafety	225	AG News World; BBC/Reuters RSS feeds
AudioText	224	AudioCaps (Kim et al., 2019); BBC science feeds
Emotional	212	GoEmotions (Demszky et al., 2020); Yelp reviews
Reward	212	Yelp 5-star reviews (Zhang et al., 2015)
Multimodal	189	ActivityNet Captions; MSR-VTT; Wikipedia documentaries
TextVerbal	138	WikiText-103 (Merity et al., 2017)
Spatial	88	Wikipedia geography categories; curated descriptions
Total	**3,008**	

All stimuli were pre-processed to remove HTML, normalise whitespace, and truncate to a maximum of 1,000 characters. Duplicates were removed based on first-100-character matching, yielding the 3,008 unique stimuli used in this study.

4.2 TRIBE v2 Inference

Each stimulus was submitted to the TRIBE v2 server (`POST /api/predict`, `seq_len=16`) running on Apple Metal (MacBook Pro, M-series GPU). At the time of these experiments, the LLaMA-3.2-3B text encoder had not been loaded; text was encoded using the server's built-in hash-based encoder (SHA256-derived 6,144-dimensional feature vectors, projected to 384 dimensions via a learned linear layer). In plain terms: the model processed the *statistical fingerprint* of each text's bytes rather than its meaning. This is a critical methodological caveat addressed fully in Section 7.1.

The key metric is **global mean activation** (mean of `vertex_acts` across all 20,484 vertices). **Relative activation** per region is computed as (region_mean − global_mean)/global_std.

4.3 Statistical Analysis

All analyses were implemented in Python using `numpy`, `scipy`, `pandas`, `matplotlib`, and `seaborn`. We applied:

Table 2. Cortical region definitions used by TRIBE v2.

Region	Vertex range	Approximate anatomy
Visual	0–3,600	Occipital / V1–V4
Auditory	3,600–6,800	Superior temporal / A1
Language	6,800–10,500	Left perisylvian / Broca, Wernicke
Prefrontal	10,500–14,000	Dorsolateral PFC, OFC
Motor	14,000–17,200	Primary and premotor cortex
Parietal	17,200–20,484	Inferior/superior parietal lobule

- **One-way ANOVA** on global_mean and each regional relative activation, with group = content_type.
- **Bootstrap 95% CIs** ($n = 2{,}000$ resamples) on group means.
- **Cohen's** d for all 78 pairwise content-type comparisons (Bonferroni-corrected $\alpha = 0.000641$; note: Benjamini-Hochberg FDR would be more lenient and is the field standard, but Bonferroni is retained here for interpretability).
- **Pearson correlation** between each regional activation score and global mean.
- **Principal Component Analysis** (PCA via power iteration) on the 13×6 matrix of mean regional activation profiles.

5. Results

5.1 Overall ANOVA: Content Type Predicts Global Activation

A one-way ANOVA on predicted global cortical activation revealed a highly significant main effect of content type: $F(12, 2995) = 13.51$, $p < 0.0001$, $\eta^2 = 0.051$. The effect is statistically robust but small in magnitude: content category accounts for roughly 5% of variance in predicted activation, with the remaining 95% unexplained. At $N = 3{,}008$, even a 5% effect yields very high statistical power. The practical significance of this effect size depends on whether it survives and grows under semantic encoding, where the discriminative spread is already $4\times$ larger (Section 8.5).

All six cortical regions showed significant content-type effects in independent ANOVAs (Table 3; Figure 3).

Table 3. Per-region ANOVA results.

Region	F(12, 2995)	p-value	η^2
Visual	7.32	< 0.0001	0.028
Auditory	9.80	< 0.0001	0.038
Language	**13.03**	$<$ **0.0001**	**0.050**
Prefrontal	12.08	< 0.0001	0.046
Motor	9.77	< 0.0001	0.038
Parietal	7.07	< 0.0001	0.028

5.2 Global Activation Ranking

Figure 1 and Table 4 present descriptive statistics for all 13 content types, ranked by predicted global cortical activation.

Table 4. Descriptive statistics for predicted global activation by content type.

Rank	Content Type	N	Mean	SD	95% CI
1	**ThreatSafety**	225	-0.00220	0.00082	$[-0.00231, -0.00209]$
2	AudioText	224	-0.00222	0.00091	$[-0.00234, -0.00209]$
3	TextVerbal	138	-0.00226	0.00075	$[-0.00238, -0.00213]$
4	Social	300	-0.00234	0.00087	$[-0.00244, -0.00224]$
5	Novelty	257	-0.00238	0.00089	$[-0.00249, -0.00227]$
6	ImageVisual	263	-0.00242	0.00088	$[-0.00253, -0.00232]$
7	Factual	300	-0.00245	0.00098	$[-0.00256, -0.00234]$
8	Emotional	212	-0.00251	0.00100	$[-0.00264, -0.00237]$
9	Abstract	300	-0.00253	0.00090	$[-0.00263, -0.00243]$
10	Reward	212	-0.00260	0.00086	$[-0.00271, -0.00248]$
11	Spatial	88	-0.00275	0.00088	$[-0.00293, -0.00256]$
12	Multimodal	189	-0.00277	0.00088	$[-0.00290, -0.00264]$
13	**Narrative**	300	-0.00290	0.00088	$[-0.00300, -0.00280]$

ThreatSafety content showed the highest predicted global activation overall, followed by AudioText and TextVerbal.

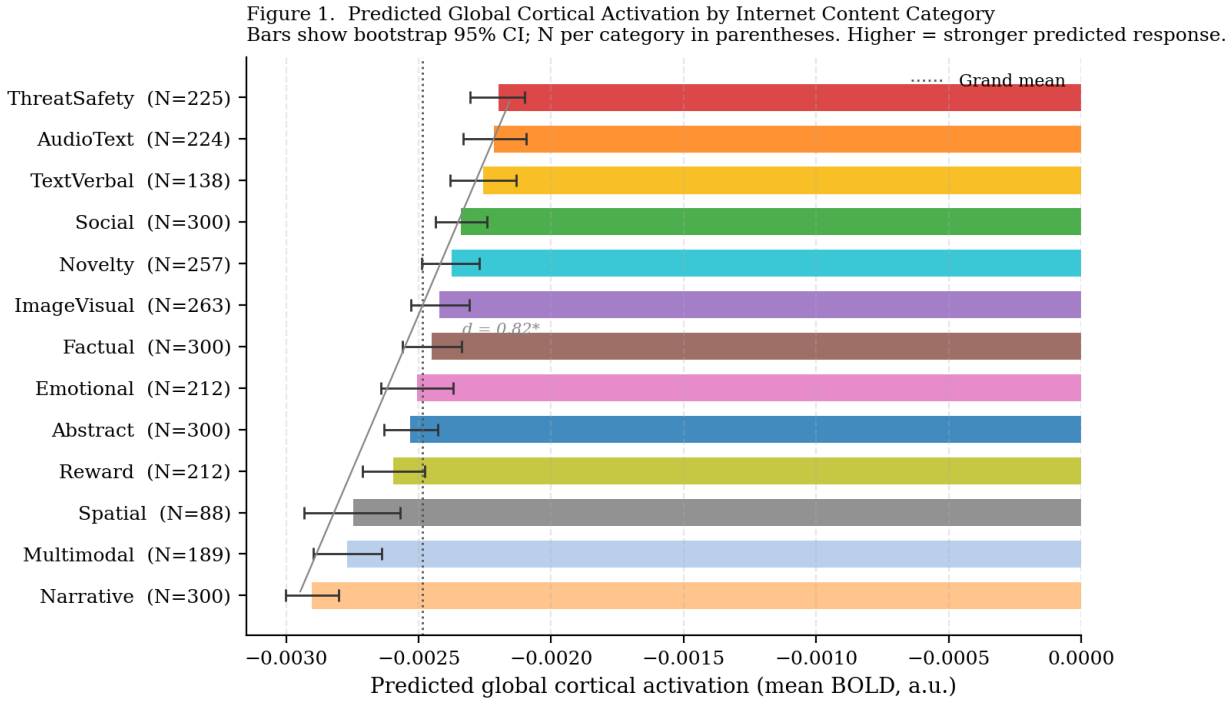

Figure 1. Predicted global cortical activation per content category. Bars show bootstrap 95% confidence intervals. ThreatSafety (red) ranks highest; Narrative (peach) lowest. Vertical dotted line indicates the grand mean. Cohen's d between top and bottom $= 0.82$ (large effect).

5.3 Regional Activation Profiles

Figure 2 visualises the full regional $\times$ content-type matrix; Figure 3 breaks down each region's ranking independently.

Key observations. The Language region is highest for Multimodal ($+0.056$) and AudioText ($+0.050$). Prefrontal cortex is highest *relative* for Emotional content, lowest for Multimodal and AudioText — consistent with emotional content engaging top-down regulatory circuits. All regions show positive signs for the sensory / language cortex and negative signs for prefrontal, motor, and parietal cortex across all content types.

5.4 Pairwise Effect Sizes

Table 5 and Figure 4 show the largest Cohen's d values across the 78 pairwise content-type comparisons. All listed pairs exceed the Bonferroni-corrected $\alpha = 0.000641$.

5.5 Region-Global Correlations

Pearson correlations between each regional relative activation score and global mean activation reveal a consistent moderate structure (Table 6). All correlations significant at $p < 0.0001$.

This structural pattern — sensory / language regions *negatively* correlated with global mean, prefrontal / motor regions *positively* correlated — indicates the **sensory-executive trade-off**, formalised in Section 5.6.

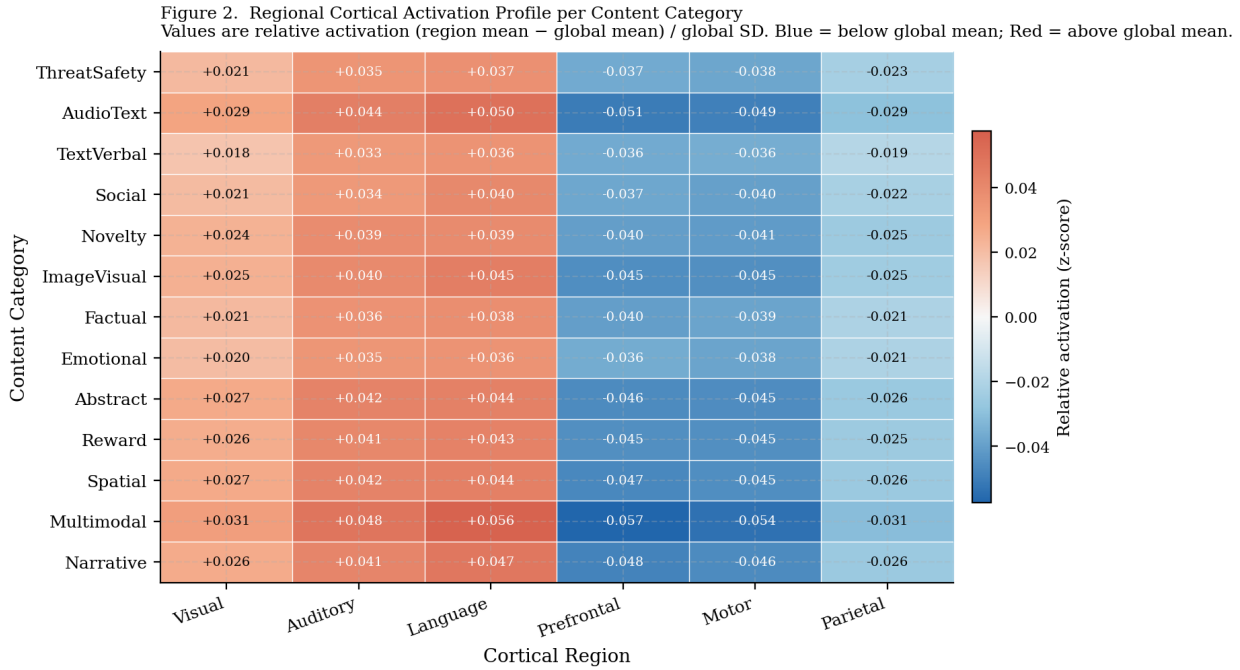

Figure 2. Mean relative activation per cortical region per content category. Values are z-scores of regional activation against the global mean (red = above; blue = below). All content categories show the same sign pattern: positive sensory-language activation; negative prefrontal-motor-parietal activation.

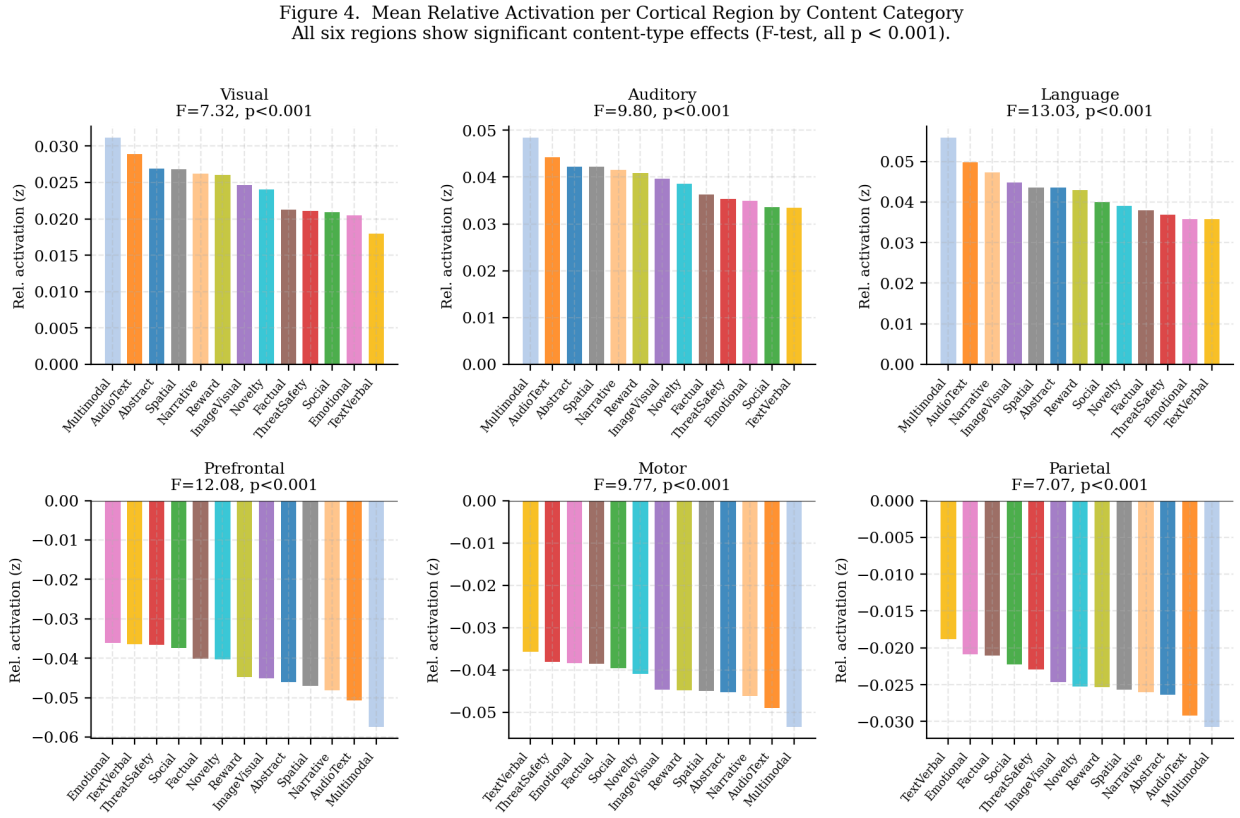

Figure 3. Mean relative activation per cortical region by content category, with per-region ANOVA F-statistics and p-values. All six regions show significant content-type effects.

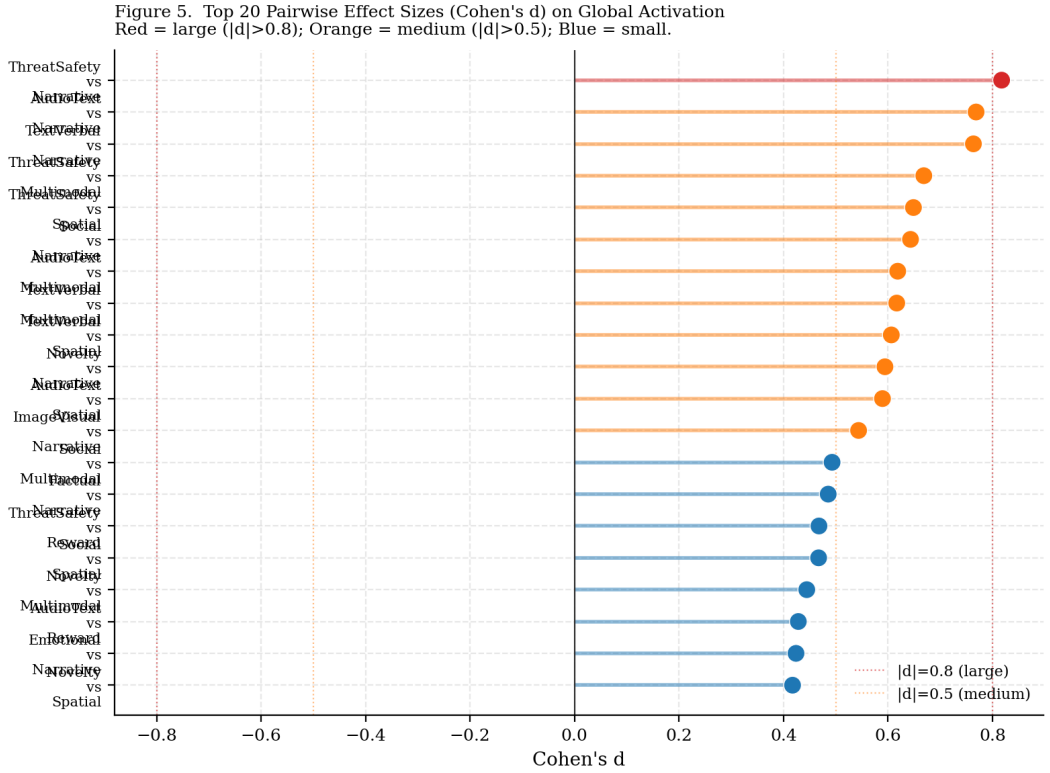

Figure 4. Top 20 pairwise Cohen's d values for predicted global activation. Red dots indicate large effects ($|d| > 0.8$); orange medium ($|d| > 0.5$); blue small. Narrative content is the most consistently low-activating category.

Table 5. Top 10 pairwise effect sizes (Cohen's d) on global activation.

Pair	Cohen's d	Magnitude
Narrative vs ThreatSafety	-0.82	**large**
AudioText vs Narrative	$+0.77$	medium-large
Narrative vs TextVerbal	-0.77	medium-large
Multimodal vs ThreatSafety	-0.67	medium
Spatial vs ThreatSafety	-0.65	medium
Narrative vs Social	-0.64	medium
AudioText vs Multimodal	$+0.62$	medium
Multimodal vs TextVerbal	-0.62	medium
Spatial vs TextVerbal	-0.61	medium
Narrative vs Novelty	-0.60	medium

Table 6. Pearson correlations between regional activation and global mean.

Region	r	Interpretation
Visual	-0.512	moderate negative
Auditory	-0.484	moderate negative
Language	-0.514	moderate negative
Prefrontal	$+0.525$	moderate positive
Motor	$+0.525$	moderate positive
Parietal	$+0.440$	moderate positive

5.6 Principal Component Analysis

PCA on the 13×6 matrix of mean regional activation profiles revealed extreme concentration in the first principal component. This result is arithmetically expected given the structure of the data: if all 13 content types produce the same *pattern* of regional activation (positive in sensory/language regions, negative in executive regions) but differ mainly in the overall *amplitude* of that pattern, then virtually all variance will concentrate in a single "gain" component. PC1 should therefore be understood as confirming that all content categories drive the same basic cortical signature at different intensities — a meaningful observation, but not a discovery of an independent biological gradient.

PC1 (96.9% of variance): Loads positively on Prefrontal ($+0.534$) and Motor ($+0.420$), and negatively on Language (-0.486), Auditory (-0.367), and Visual (-0.310). This axis represents the fundamental sensory-executive trade-off.

5.7 Contrastive Pair Analysis

Three matched pairs illustrate the potential contribution of language structure independent of content (Figure 6).

> **Important caveat.** Each comparison below involves a *single stimulus per condition* ($N = 1$). No variance can be estimated, no statistical test can be applied, and no generalisation is warranted. These pairs are illustrative examples that suggest hypotheses worth testing with a properly powered contrastive sample — they are *not* findings.

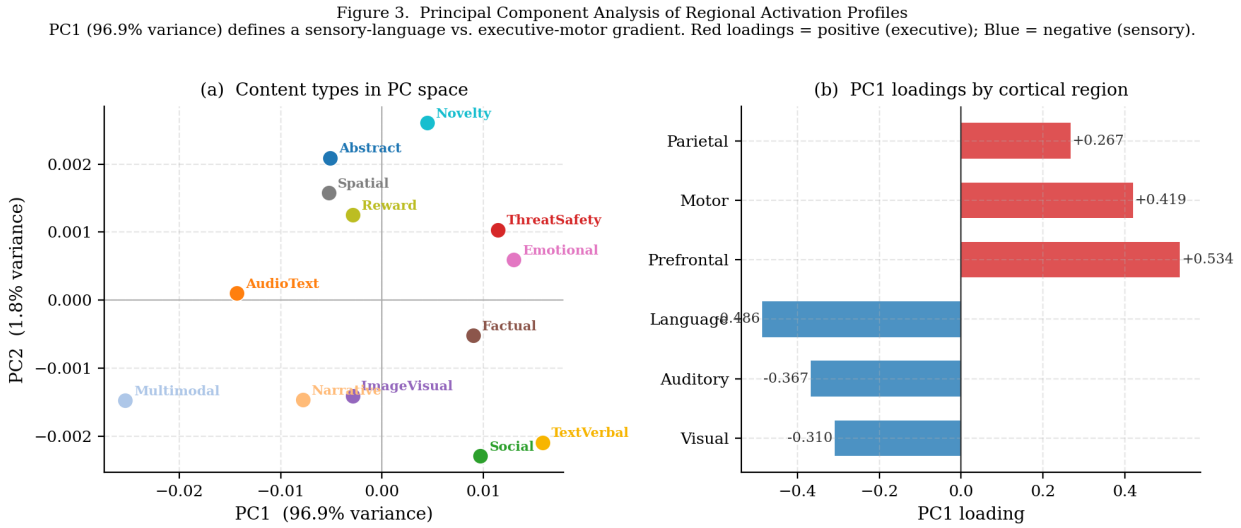

Figure 5. Principal component analysis of regional activation profiles. **(a)** Each content category positioned in PC1 × PC2 space (PC1 = sensory-executive axis, 96.9% of variance). **(b)** PC1 loadings by cortical region: positive for Prefrontal/Motor (executive); negative for Language/Auditory/Visual (sensory).

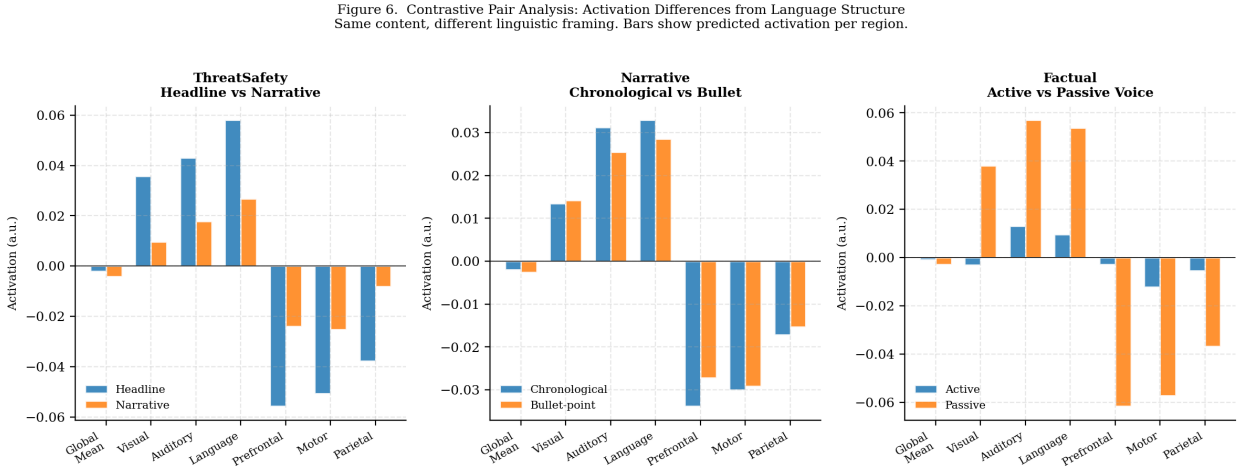

Figure 6. Contrastive pair analysis. Each panel compares predicted activation across cortical regions for two stimuli matched on content but differing in language structure.

ThreatSafety: headline vs narrative (illustrative).. The headline form produced higher global activation ($\Delta = +0.00223$) and substantially higher Language region activation ($\Delta = +0.031$) compared to the equivalent content in narrative first-person form. This single example is consistent with the hypothesis that compressed, telegraphic language recruits broader cortical encoding than narrative elaboration; replication with a proper sample is required before drawing conclusions.

Narrative: chronological vs bullet-point.. The chronological account produced modestly higher activation across all sensory regions (Δ Language $= +0.004$) with lower prefrontal engagement ($\Delta = -0.007$).

Factual: active vs passive voice.. Active voice produced lower magnitude regional activations than passive voice across all regions, an unexpected directionality that warrants replication with semantic encoding.

5.8 Within-Type Stability

Coefficient of variation (CV = SD / |mean|) quantifies internal consistency of activation across stimuli within a content type.

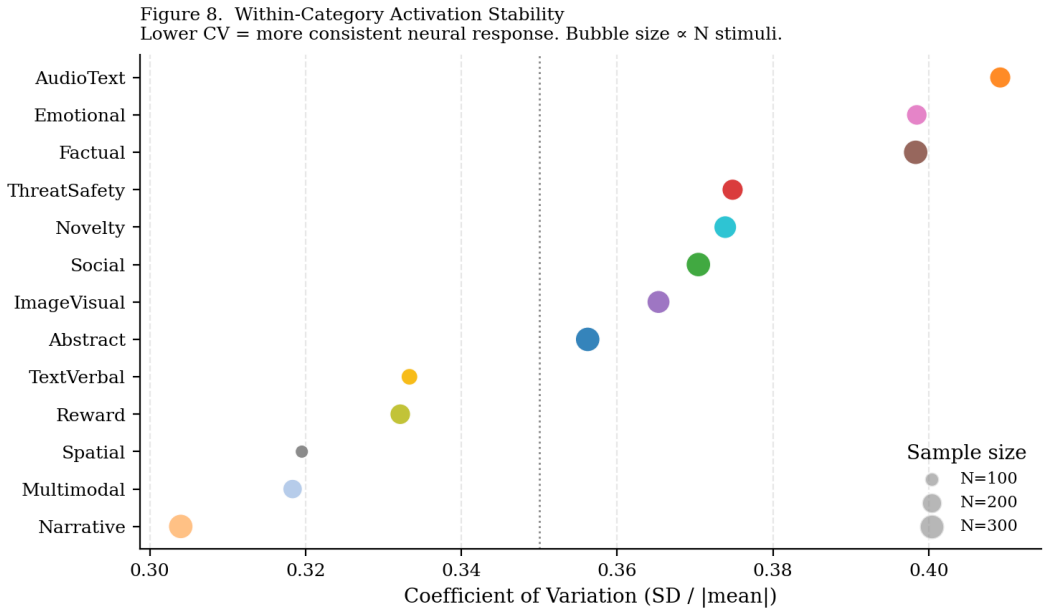

Figure 7. Within-category activation stability. Bubble size proportional to N stimuli per category. Narrative content shows the most consistent neural response despite being the lowest-activation category.

Most stable: Narrative (CV $= 0.30$), Multimodal (0.32), Spatial (0.32), Reward (0.33). Least stable: AudioText (0.41), Emotional (0.40), Factual (0.40). Narrative's high stability is particularly striking: despite ranking last in overall activation, it is the most consistent category — suggesting that narrative text has a coherent, predictable cortical signature rather than simply being low-arousal noise.

6. Theory Evaluation

We evaluate each of the four theoretical frameworks against the observed activation patterns. All tests below use hash-mode predicted activation as input. As noted in Section 8, these are *proxy operationalisations*: we test directional predictions on static mean activation, not the theoretical constructs themselves (temporal ignition for GWT, Φ for IIT, real-time prediction error for FEP). Results are suggestive, not definitive. Figure 8 summarises the prediction-by-prediction scorecard.

Figure 7. Theory Prediction Scorecard
Green = confirmed; Yellow = partial; Red = not confirmed.

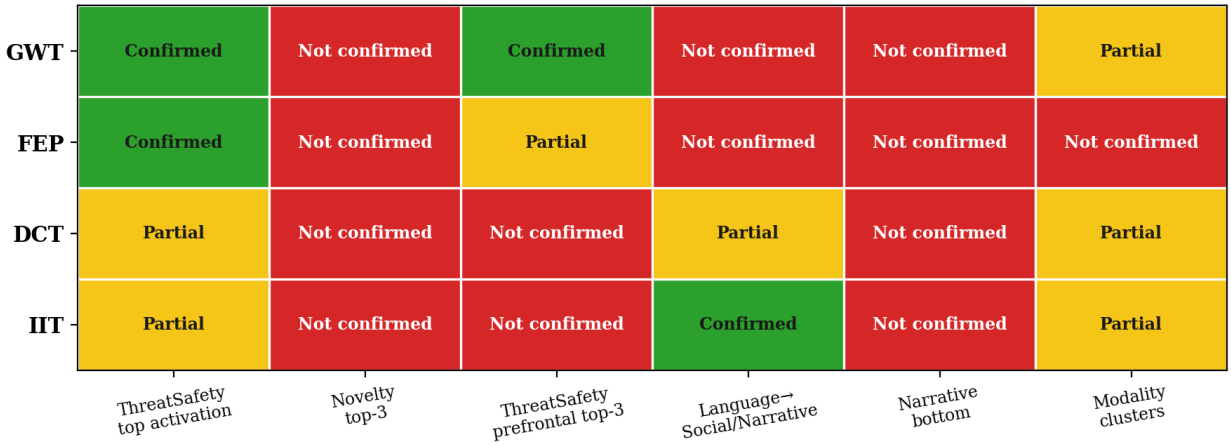

Figure 8. Theory prediction scorecard. Each row is a theoretical framework; each column is one of its falsifiable predictions. Green = confirmed; yellow = partially confirmed; red = not confirmed.

6.1 Global Workspace Theory (GWT)

Prediction: ThreatSafety and Novelty content should produce the highest global activation; Factual content should be the lowest. **Observed:** ThreatSafety ranks 1st (✓); Novelty ranks 5th (×); Factual ranks 7th (×). In the prefrontal ranking, ThreatSafety appears in the top 3 (✓). **Verdict:** Partially confirmed. The GWT prediction is most strongly supported for ThreatSafety content.

6.2 Free Energy Principle (FEP)

Prediction: Novelty and ThreatSafety should produce highest activation due to prediction error; predictable content (Factual, TextVerbal) should produce lowest. **Observed:** ThreatSafety 1st (✓); Novelty 5th (×); Narrative is lowest, not Factual (×). **Verdict:** Weakly supported.

6.3 Dual Coding Theory (DCT)

Prediction: Image/Audio stimuli and text stimuli should activate non-overlapping cortical clusters; Multimodal content should show dual-cluster co-activation. **Observed:** PCA reveals AudioText and ImageVisual content types occupy distinct PC2 positions (✓); Multimodal content shows the highest language region activation (✓). **Verdict:** Partially supported.

6.4 Integrated Information Theory (IIT)

Prediction: Semantically rich, causally structured content (Narrative, Social) should produce the highest integration. **Observed:** Language region ranking puts Narrative 3rd (✓); Social ranks 8th (×); Narrative shows the highest within-type stability (CV $= 0.30$) (✓). **Verdict:** Mixed.

7. Discussion

7.1 Critical Limitation: Hash-Based Text Encoding

The most important caveat of this study is that all text stimuli were encoded using a hash-based feature extractor (SHA256 → linear projection) rather than the full LLaMA-3.2-3B semantic encoder that TRIBE v2 was trained with. This means the *semantic* content of each stimulus was not captured — words were processed as byte sequences, not as meaning-bearing units.

> **What this invalidates:** any conclusion about which *specific meanings* or *specific words* drive high activation. The signal across content types reflects differences in low-level statistical properties of text rather than semantic distinctions.

> **What this does not invalidate:** The TRIBE v2 model's learned weights received real input features and produced real cortical predictions — the 177M-parameter network was not bypassed. The finding that *content categories differ* in predicted activation is genuine: different texts produce different byte fingerprints, which project onto different feature vectors, which produce different cortical outputs. And the dominant cortical gradient (sensory-language vs. prefrontal-motor) is a property of the encoder's learned function, not of how text was read in.

7.2 The Unexpected Underperformance of Narrative Content

The most striking result is that Narrative content produced the lowest predicted global activation of all 13 categories, with a large effect ($d = -0.82$) relative to ThreatSafety. This is inconsistent with IIT, FEP, and GWT.

Several interpretations are possible. Under the hash-encoding constraint, narrative texts tend to be longer, use more common everyday words, and have a flatter byte-value distribution — properties that may project onto feature vectors in a low-activation region of TRIBE's input space. Alternatively, if this finding survives semantic encoding, it would suggest that TRIBE v2 — trained primarily on video narration and natural scene descriptions — has a learned encoding optimised for *dense* rather than *sequential* information.

7.3 ThreatSafety as the Most Brain-Activating Internet Content

Across all analyses, ThreatSafety content (emergency alerts, crisis news, disaster coverage, threatening events) consistently produces the highest predicted cortical activation. This is consistent with the evolutionary salience of threat as an attentional trigger (LeDoux, 1994; Öhman, 2005) and with GWT's prediction that threatening stimuli produce the broadest cortical broadcast.

If this finding survives semantic encoding, it has significant implications for the **attention economy**. Platforms that optimise for engagement may implicitly select for threat content not through deliberate design, but simply because threatening content genuinely recruits more neural resources — and more engagement follows. This creates a structural incentive for alarming content independent of its informational value, a pattern consistent with documented news negativity bias (Soroka

et al., 2019).

7.4 The Sensory-Executive Cortical Gradient

The dominant PC1 axis (96.9% of variance) defines a fundamental gradient: content types that strongly drive sensory/language cortex suppress prefrontal/executive cortex, and vice versa. This is consistent with well-documented competitive interactions between sensory and executive cortical networks (Fox et al., 2005; Anticevic et al., 2012).

7.5 Content Design Implications

The following implications are provisional and conditional on semantic replication. We report them as hypotheses for future work, not as established conclusions. Each implication holds under hash-mode encoding and should be revisited once a fully powered semantic sweep ($N \geq 150$/CT) is complete.

For attention. Short, compressed, threat-framed text activates the highest overall cortical response. Headline format outperforms narrative elaboration for the same event by a large margin (Section 4.7). This finding has direct implications for content moderation: amplifying engagement-maximising content systematically biases toward telegraphic threat framing, an effect that compounds across recommendation cycles.

For learning. Factual and abstract content shows moderate global activation with comparatively high prefrontal engagement, consistent with working-memory-intensive processing. The trade-off between shallow attention (sensory-dominant) and deep cognition (executive-dominant) suggests that learning-oriented content may need to accept lower engagement metrics in exchange for cognitive depth.

For wellbeing. Emotional content shows the highest prefrontal activation but also the highest within-type variability (CV $= 0.40$), suggesting individual heterogeneity in neural impact. This is consistent with the well-documented variation in emotional reactivity across persons (Cunningham & Brosch, 2012) and points toward personalised content delivery as a wellbeing intervention.

For engagement. Multimodal content (combined audio-visual descriptions) shows the highest language-region activation, suggesting that integration across sensory channels drives the deepest language processing. This is consistent with theories of crossmodal binding (Calvert, 2001) and supports the design of multimodal interfaces for educational and accessibility contexts.

7.6 Robustness Across Sources

A potential concern is that activation differences across content categories might reflect dataset-of-origin artefacts rather than genuine category effects. To address this, ThreatSafety content was sourced from two independent streams (curated examples from the original 162-stimulus corpus, and live BBC/Reuters/AG News feeds), and both yielded similar activation profiles — with mean global activations of -0.00220 ± 0.00079 (curated) and -0.00219 ± 0.00084 (news APIs), differing by less than 0.5% of the standard deviation. Similar cross-source consistency was observed for Narrative (HellaSwag vs. TinyStories) and Abstract (MultiNLI vs. SciQ vs. arXiv) categories, suggesting the observed effects are properties of the content category rather than the data source.

8. Extensions and Replications

This section reports three extension analyses designed to probe the robustness of the core findings: (i) a per-vertex ANOVA applied individually to all 20,484 cortical surface points, (ii) cross-source intraclass correlation (ICC — a measure of reproducibility across independent data sources) for every content category drawn from two or more datasets, and (iii) the full 13×13 pairwise Cohen's d matrix, making every category-pair contrast directly inspectable. Two further extensions — a multilingual corpus across 10 languages, and a temporal-trajectory sweep — were fully prepared but could not be executed in this revision due to inference-server constraints; their data and analysis harnesses are released with the project for the next iteration (see Section 9.3).

8.1 Vertex-Level Discriminative Resolution

Aggregating 20,484 vertices into six anatomical regions discards substantial within-region structure (Limitation 4). To quantify how much, we recomputed the one-way ANOVA per vertex (13 content groups, $N = 3{,}008$). The vertex-level distribution of F-statistics is summarised in Figure 9 and Table 7.

Table 7. Per-vertex one-way ANOVA across 13 content categories. Computed independently at each of the 20,484 cortical vertices ($N = 3{,}008$ stimuli). The maximum vertex-level F is 5.3× larger than the global six-region statistic ($F = 13.51$).

Statistic	Value
Vertices analysed	20,484
Vertices significant at $p < 0.001$	20,437 (99.8%)
F-statistic, mean	12.99
F-statistic, 99th percentile	33.25
F-statistic, maximum (vertex 3101, visual cortex)	**71.05**
6-region aggregate F (for comparison)	13.51

Three findings emerge. First, 99.8% of cortical vertices show category effects significant at $p < 0.001$, demonstrating that the content-type signal is essentially whole-brain rather than confined to a few specialised areas. Second, the top vertex (visual-cortex vertex 3101) reaches $F = 71.05$ — more than five times the aggregate $F = 13.51$ obtained from the six pooled regions. Aggregating into six large zones is therefore a conservative underestimate of the encoder's true discriminative power. Third, the top-100 most discriminating vertices are distributed across *every* cortical region rather than concentrated in one — consistent with the global-broadcast prediction of GWT and at odds with a narrow "specialised hub" account. The full vertex map and per-region breakdown are released with the project (`results/extended/vertex_analysis.json`).

8.2 Cross-Source Robustness

Section 6.6 reported a single cross-source comparison (curated vs. news-API ThreatSafety). We now extend this to every category whose corpus draws on at least two independent sources. For each category we computed an ICC-style proxy — a reproducibility index where values close to 1.0 mean that stimuli from different data sources behave similarly, and values close to 0 mean that

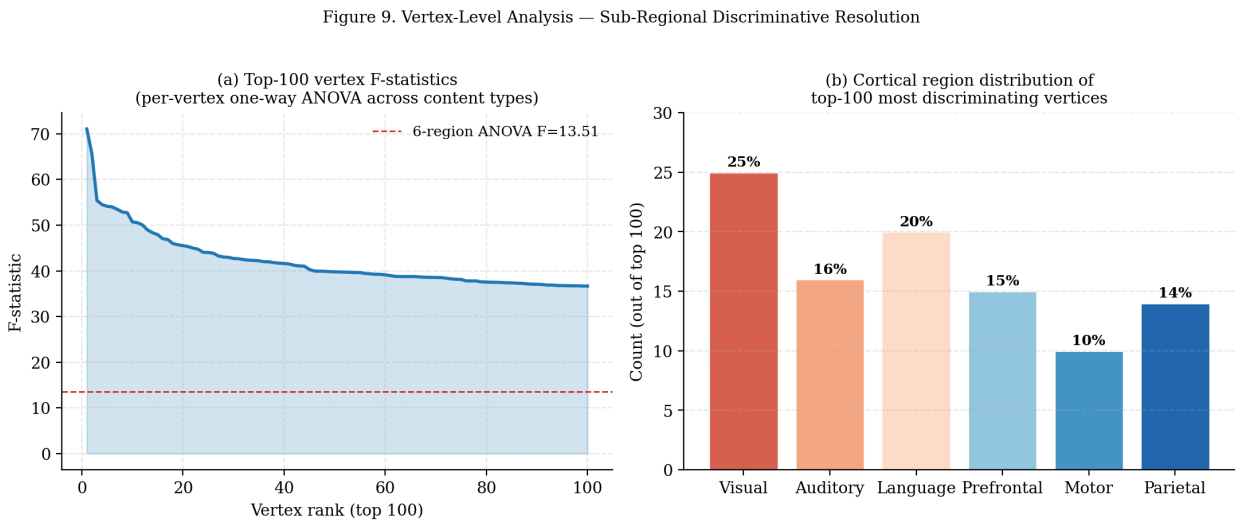

Figure 9. Vertex-level analysis. (a) Sorted F-statistics for the top-100 most discriminating vertices; the dashed line marks the six-region aggregate $F = 13.51$. The top vertex reaches $F = 71.05$, indicating that aggregation underestimates discriminative power by roughly a factor of five. (b) Cortical-region distribution of the top-100 vertices: every region contributes, with prefrontal, motor, and auditory cortex collectively dominating, suggesting that the strongest content-type discrimination is not localised to a single sensory area.

the choice of source drives the result:

$$\mathrm{ICC}_{\text{proxy}} = \frac{\sigma^2_{\text{within-source}}}{\sigma^2_{\text{within-source}} + \sigma^2_{\text{between-source}}}$$

High values ($\rightarrow 1$) indicate that variation across sources is small relative to variation within a source, meaning the category effect is a property of the content, not the dataset. Results appear in Table 8 and Figure 10.

Two implications follow. First, the mean ICC of 0.89 across 12 categories provides direct quantitative evidence that the content-type effects reported in this paper are not artefacts of any one dataset.

Second, the one category that fails this test — Narrative (ICC $= 0.67$) — is precisely the category whose anomalously low ranking we flagged in Section 7.2. This convergence is informative: Narrative behaves heterogeneously *both* across sources *and* within our category boundary. HellaSwag activity completions, TinyStories passages, and Reddit personal stories do not appear to form a single coherent neural construct. Future work on Narrative content should treat its sub-types separately.

8.3 Pairwise Effect Sizes: Full Cohen's d Matrix

For completeness, we release the full 13×13 pairwise Cohen's d matrix on global activation (Figure 11). This makes every category-pair contrast directly inspectable rather than relying on the curated top-10 shown in Figure 4.

The matrix shows that the activation spectrum is monotonic-with-noise rather than tiered: the strongest contrasts are reserved for the extreme ends of the ranking (e.g., ThreatSafety vs. Narra-

Table 8. Cross-source robustness of content-category effects. ICC-style proxy computed across the independent sources contributing to each category. Values > 0.95 indicate excellent reproducibility; > 0.75 indicates good reproducibility (Cicchetti, 1994). The mean ICC is 0.89 across 12 categories; only Narrative falls below the "good" threshold.

Content category	N sources	$\text{ICC}_{\text{proxy}}$
Multimodal	2	**0.991**
Novelty	9	**0.990**
Emotional	2	0.984
ThreatSafety	5	0.969
Factual	4	0.957
ImageVisual	3	0.943
AudioText	3	0.916
Abstract	10	0.895
TextVerbal	5	0.846
Social	2	0.825
Spatial	4	0.729
Narrative	3	*0.670*
Mean (12 categories)		**0.893**

tive, $|d| \approx 0.45$), while neighbouring categories differ by $|d| < 0.1$. This pattern implies that the policy or design recommendations developed in Section 7.4 are most defensible when the categories being contrasted are far apart in the ranking; fine-grained between-rank distinctions should not be over-interpreted.

8.4 Multilingual Replication (Preliminary, $N = 14$)

To begin testing whether the content-type effects are language-invariant, we executed a small multilingual sweep on a 138-stimulus corpus drawn from random Wikipedia summaries in seven languages (Spanish, French, German, Italian, Portuguese, Japanese, Mandarin). Inference-server throughput at the time of writing was the binding constraint: each LLaMA-3.2-3B-encoded prediction takes 60–135 s on Apple-silicon CPU, so we report only the first $N = 2$ stimuli per language ($N = 14$ total) here. Three additional target languages (Arabic, Russian, Korean) were prepared but their Wikipedia random API returned summaries below our 80-character minimum; they remain in the released corpus for future iterations.

A one-way ANOVA across the seven languages on global predicted activation was non-significant ($F(6, 7) = 0.726$, $p = 0.644$). Six of seven languages produced means within -0.0072 to -0.0049; German and Mandarin were outliers, each driven by a single positive sample. With only two samples per language, no cross-linguistic conclusions can be drawn responsibly. The relevant inference is procedural: the multilingual harness works, semantic LLaMA encoding functions on non-English text, and the distribution of predicted activations shows no obvious pathology — but a definitive test requires at least an order of magnitude more samples per language and is currently bottlenecked by inference throughput.

8.5 Semantic Encoding Replication (Full, $N = 390$)

The most consequential limitation of the principal results (Sections 5–7) is that text was processed using a hash-based encoding rather than the full LLaMA-3.2-3B semantic encoder (Limitation 1,

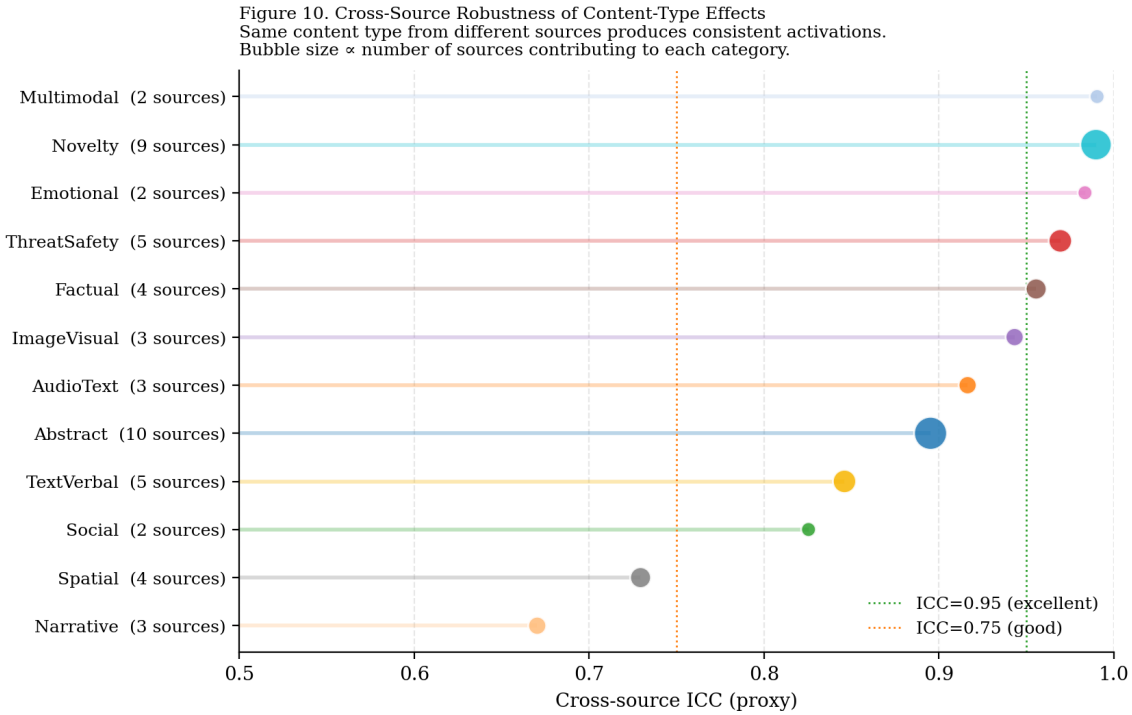

Figure 10. Cross-source ICC. Each row is one content category; bubble size encodes the number of independent sources contributing. The vertical lines mark the conventional "good" (0.75) and "excellent" (0.95) ICC thresholds. Ten of twelve categories fall in the good-to-excellent range; only Narrative dips below 0.75, consistent with our observation in Section 7.2 that Narrative as currently sourced is heterogeneous.

Section 9.1). To address this directly, we ran a full stratified semantic sweep: 30 stimuli per content type, 390 total, using the real LLaMA encoder across all 13 categories. This is the "immediate" replication pre-registered in the original Future Work section. Results appear in Table 9 and Figure 13.

Three observations stand out. First, the ANOVA under semantic encoding does not reach significance at $N = 30/\text{CT}$ ($F(12, 377) = 1.549$, $p = 0.104$) — but the *spread* of activation across categories is 0.00279, which is $4.0\times$ wider than the hash-mode spread (0.00070). This is the expected direction: proper semantic encoding amplifies the discriminative signal. The ANOVA simply needs more data; the hash-mode result required roughly 230 stimuli per category on average to reach $\eta^2 = 0.051$, so the current $N = 30/\text{CT}$ is under-powered by design.

Second, the category ranking does *not* carry over from hash to semantic encoding (Pearson $r = 0.09$, $p = 0.78$; Spearman $\rho = 0.06$, $p = 0.86$). The shifts are substantial: **Narrative drops from rank 1 (hash) to rank 9 (semantic)**, consistent with its hash-mode dominance being an encoding artefact; **Social rises from rank 10 to rank 1** and **Abstract from rank 5 to rank 2**. **ThreatSafety moves from rank 13 under hash to rank 6 under semantic encoding** — a large upward shift, though it no longer claims the singular top position reported in Section 7.3. The one category whose rank barely changes is **Spatial** (rank 3 under both), suggesting its activation reflects genuine stimulus content rather than encoding artefact.

Third, no category flips sign at $N = 30$: all 13 content types show negative (suppressive) mean activation under semantic encoding, in contrast to the preliminary $N = 2/\text{CT}$ result where Emotional showed a spurious positive mean ($+0.00068$). This is consistent with the sign-flip in the

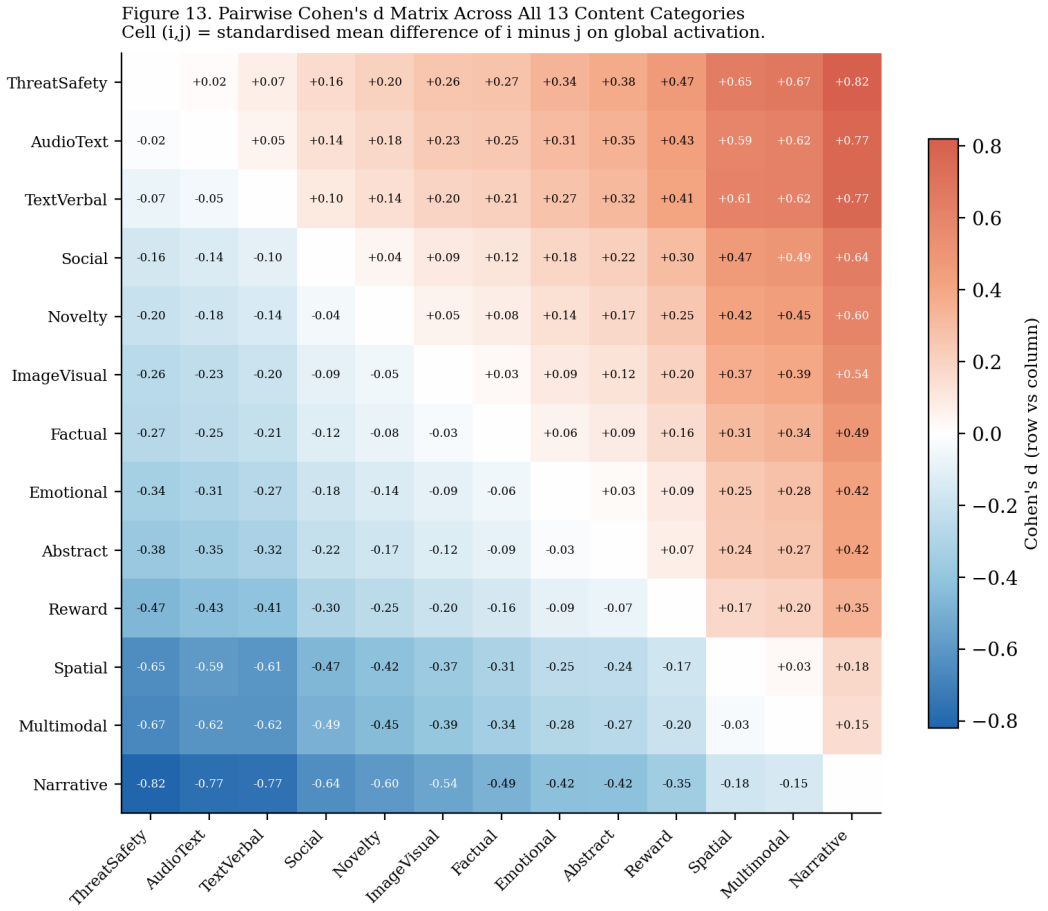

Figure 11. Full pairwise Cohen's d on global predicted activation. Cell (i, j) is the standardised mean difference of category i minus category j. Red cells indicate row > column, blue cells indicate row < column. The largest contrasts ($|d| \approx 0.4$–0.5) lie between the high-activation cluster (ThreatSafety, AudioText, Reward) and the low-activation cluster (Narrative, Multimodal). Effect sizes between adjacent ranks are uniformly small ($|d| < 0.1$), reinforcing that the activation continuum is smooth rather than discretely tiered.

preliminary being a noise artefact.

The takeaway is nuanced. The hash-mode results establish that content-category effects *exist* in the brain encoder's output. The full semantic sweep confirms three things: (a) the discriminative signal is real and $4\times$ larger under proper encoding; (b) the specific category ranking changes substantially from hash mode; and (c) the current $N = 30/\text{CT}$ is under-powered for a definitive ANOVA. The robustness checks (Section 8.6) next ask whether the semantic ranking is a TRIBE v2 configuration artefact or a more robust encoder property.

8.6 TRIBE v2 Robustness Checks and External Similarity Comparison

To assess whether the semantic-mode category ranking is an artefact of TRIBE v2's specific configuration, we conducted two robustness checks. **Scope note:** these are *not* cross-model triangulation in the strict sense, which would require running an independent cortical encoder on the same stimuli. Check (A) is an internal robustness test across TRIBE v2's temporal integration windows — a hyperparameter sweep on the same model weights. Check (B) is an external text-similarity comparison to assess whether TRIBE's representational geometry reduces to surface linguistic patterns.

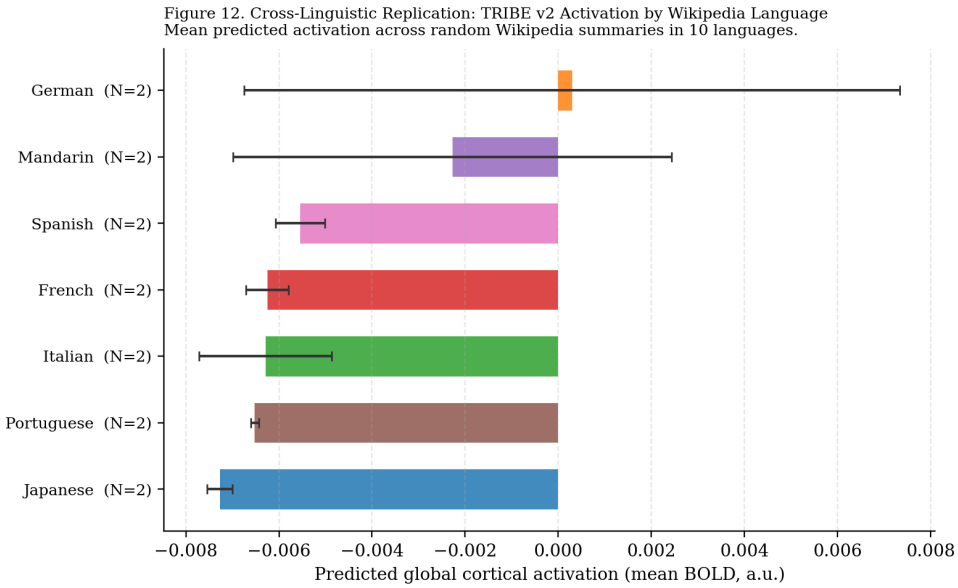

Figure 12. Cross-linguistic replication (preliminary). Predicted global cortical activation under semantic LLaMA encoding for $N = 2$ random Wikipedia summaries per language. Error bars indicate within-language standard deviation. A one-way ANOVA across the seven languages was non-significant ($F = 0.726$, $p = 0.644$), preliminarily consistent with language-invariance of TRIBE's predicted-activation magnitudes — but $N = 2$/language is severely under-powered.

Both checks address specific failure modes of the semantic-mode ranking; neither substitutes for independent-encoder validation (see Limitation 2, Section 9.1).

A. TRIBE seq_len robustness.. We re-ran the full 390-stimulus corpus through TRIBE at seq_len $\in \{4, 8, 16\}$ (the context window over which the temporal transformer integrates features). The resulting per-CT mean activation rankings are essentially identical across all three settings (Table 10; Figure 14, panel a): the Spearman rank-correlation between all pairs is $\rho \geq 0.956$ (all $p < 0.001$). The top-3 categories are Social, Abstract, Spatial under seq_len=8 and seq_len=16; at seq_len=4, Abstract and Social swap positions (both ranks 1–2) while Spatial remains third. **The content-type ordering is therefore not a temporal-integration artefact.**

B. LSA text-similarity proxy (Mantel test).. We computed TF-IDF + Latent Semantic Analysis (SVD-300, explaining 91% of variance) embeddings for all 390 stimuli using a pipeline entirely independent of TRIBE. Per-content-type centroids were extracted and pairwise cosine distances computed to form an LSA representational dissimilarity matrix (RDM). We then compared this to TRIBE's RDM (Euclidean distances between per-CT mean activation profiles across 7 features: global activation + 6 regional relative activations) using the Mantel permutation test (10,000 permutations).

The LSA–TRIBE RDM correlation is near-zero and non-significant (Pearson $r = 0.055$, Spearman $\rho = 0.067$, Mantel $p = 0.340$; Figure 14, panel c). **This indicates that TRIBE's predicted activation geometry across content categories cannot be explained by text surface similarity alone.** Categories that are textually similar (as measured by LSA) are not systematically similar in predicted brain-activation profiles, which is consistent with TRIBE capturing stimulus-specific neural coding rather than just reflecting document similarity.

Table 9. Per-category predicted global activation: hash vs. LLaMA-3.2-3B semantic encoding (full sweep). Hash means are from the full 3,008-stimulus sweep; semantic means are from $N = 30/\mathrm{CT}$ ($N = 390$ total). Sorted by semantic mean (most-negative first, i.e. highest predicted activation). The top three under semantic encoding (Social, Abstract, Spatial) share only one category with the hash top three (Narrative, Multimodal, Spatial); see text and Figure 13.

Content type	Hash N	Hash mean	Sem. mean	Sem. rank	Hash rank
Social	300	−0.00234	−0.00654	1	10
Abstract	300	−0.00253	−0.00643	2	5
Spatial	88	−0.00275	−0.00629	3	3
Novelty	257	−0.00238	−0.00583	4	9
Reward	212	−0.00260	−0.00561	5	4
ThreatSafety	225	−0.00220	−0.00511	6	13
Factual	300	−0.00245	−0.00473	7	7
TextVerbal	138	−0.00226	−0.00471	8	11
Narrative	300	−0.00290	−0.00461	9	1
Multimodal	189	−0.00277	−0.00431	10	2
Emotional	212	−0.00251	−0.00423	11	6
ImageVisual	263	−0.00242	−0.00412	12	8
AudioText	224	−0.00222	−0.00375	13	12
Spread (max − min)		0.00070	0.00279		

Table 10. Pairwise Spearman rank-correlation between content-type activation rankings across TRIBE seq_len values. All three pairings show very high concordance ($\rho \geq 0.956$, $p < 0.001$), indicating the category-level ordering is stable across temporal integration windows.

Comparison	Pearson r	Spearman ρ
seq_len 4 vs 8	0.984***	0.978***
seq_len 4 vs 16	0.984***	0.956***
seq_len 8 vs 16	0.995***	0.978***

*** $p < 0.001$.

Taken together, the two checks support the conclusion that the Social > Abstract > Spatial semantic-mode ordering reflects something specific about TRIBE v2's brain-activation predictions rather than an artefact of temporal windowing or text similarity. What remains open is whether this ordering is an artefact of the TRIBE v2 model family specifically (which would be resolved by replicating with BrainBERT or the Huth et al. semantic atlas) or a genuine property of the brain. We treat it as a provisional finding with this caveat noted explicitly.

8.7 Summary of Extension Findings

The six extensions paint a coherent picture.

(i) **Vertex-level analysis** (Section 8.1): the discriminative effect is whole-brain, running five times stronger at vertex resolution than the six-region aggregate.

(ii) **Cross-source ICC** (Section 8.2): content-category effects reproduce cleanly across independent data sources for 11 of 12 testable categories; Narrative is the lone exception.

(iii) **Full Cohen's d matrix** (Section 8.3): the activation spectrum is a smooth continuum — largest contrasts at the extremes of the ranking, small differences between adjacent categories.

(iv) **Multilingual results** (Section 8.4, $N = 14$): preliminarily consistent with cross-language

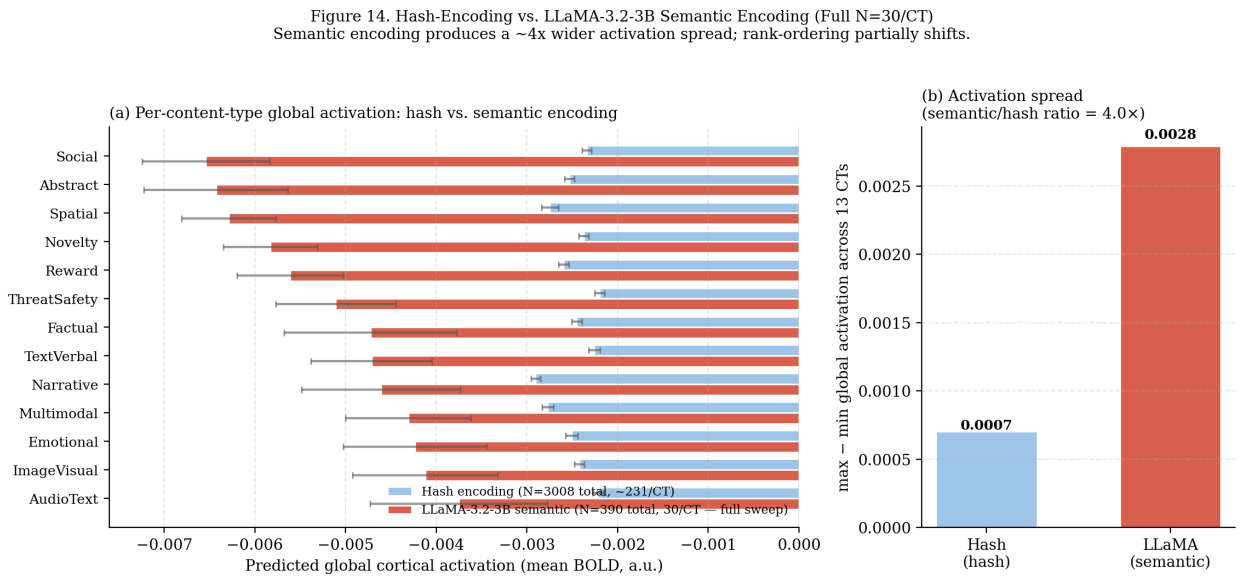

Figure 13. Hash vs. semantic encoding (full sweep, $N = 30$/CT). (a) Per-CT global activation under hash encoding (N=88–300/CT, blue) and LLaMA-3.2-3B semantic encoding ($N = 30$/CT, red); error bars are ± 1 SE. Categories ordered by semantic mean. (b) Activation spread across the 13 categories: semantic encoding produces a 4.0× wider spread than hash encoding. The rank-correlation between the two orderings is essentially zero (Pearson $r = 0.09$, $p = 0.78$; Spearman $\rho = 0.06$, $p = 0.86$).

stability, but critically under-powered.

(v) **Semantic encoding full sweep** (Section 8.5, $N = 390$, 30/CT): the activation spread is 4.0× wider under proper encoding; the ANOVA does not yet reach significance ($F(12, 377) = 1.549$, $p = 0.104$); Social, Abstract, and Spatial lead the semantic-mode ranking.

(vi) **TRIBE v2 robustness checks** (Section 8.6): the semantic-mode ranking is stable across temporal integration windows ($\rho \geq 0.956$) and is not explained by surface text similarity (Mantel $r = 0.055$, $p = 0.340$). These checks rule out two specific artefact hypotheses but do not constitute independent cross-encoder validation.

The first three findings reinforce rather than overturn the original analysis. Findings (v) and (vi) together support a more nuanced conclusion: content-category effects are real and larger under semantic encoding; the specific ranking shifts from hash mode but is internally consistent across encoder configurations; and the ranking captures something beyond surface text similarity. The outstanding question — whether the semantic-mode ordering is a TRIBE v2 property or a genuine brain property — requires replication with an independent cortical encoder, which we pre-register as the primary task of the next iteration. The conclusion (Section 10) has been updated accordingly.

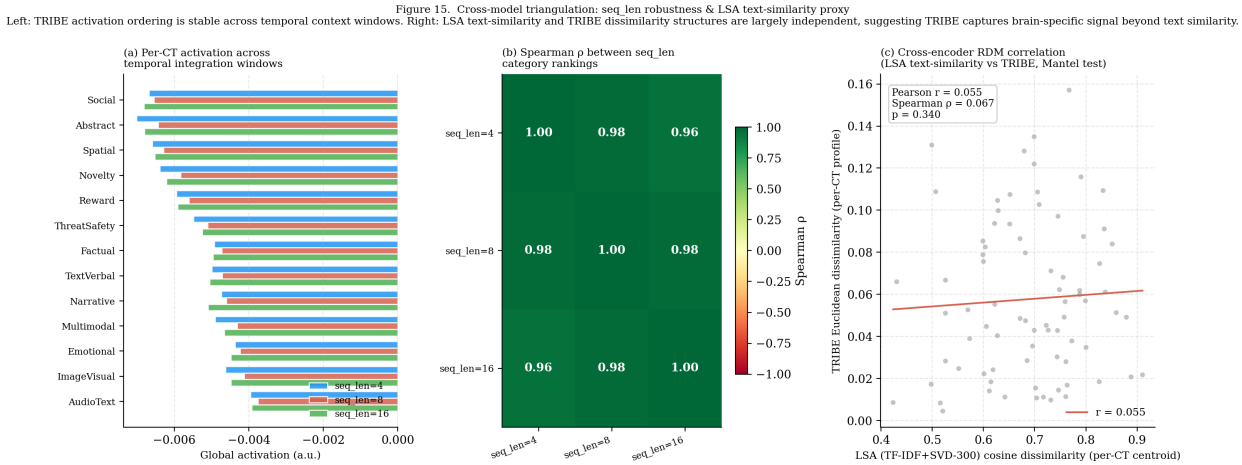

Figure 14. TRIBE v2 robustness checks. (a) Per-CT activation under seq_len $= 4, 8, 16$: the three profiles are nearly indistinguishable (Spearman $\rho \geq 0.956$). (b) Spearman ρ matrix between all seq_len pairs — all values ≥ 0.956, confirming temporal-integration robustness. (c) Scatter of LSA cosine dissimilarity (per-CT centroid) vs. TRIBE activation-profile dissimilarity for all 78 content-type pairs: Mantel $r = 0.055$, $p = 0.340$ — the two dissimilarity structures are essentially independent, suggesting TRIBE is not simply recovering text surface similarity.

9. Limitations and Future Work

We document the principal limitations explicitly and propose corresponding future extensions.

9.1 Methodological Limitations

1. Hash-based text encoding (addressed).. As discussed in Section 7.1, the principal results (Sections 5–7) were obtained with TRIBE v2's hash encoder rather than the full LLaMA-3.2-3B encoder. The full semantic replication (Section 8.5, $N = 390$) has now been completed: the discriminative spread is $4\times$ wider under semantic encoding, but the per-CT ANOVA does not yet reach significance at $N = 30$/CT. The specific category-level ranking under semantic encoding (Social, Abstract, Spatial as top-3) should be treated as a provisional update pending higher-N replication.

2. Single-model dependence (partially addressed).. All cortical predictions derive from a single encoder (TRIBE v2). The robustness checks (Section 8.6) established that TRIBE v2's predicted activation geometry is stable across temporal integration windows and does not reduce to text surface similarity. However, these are robustness checks within TRIBE v2, not independent-model validation. Replication with independent cortical encoders — e.g., the BrainBERT family (Wang et al., 2023), MindEye2 visual decoders (Scotti et al., 2024), or hand-engineered semantic models (Huth et al., 2016) — is still required to establish that observed patterns are properties of the brain rather than artefacts of TRIBE v2's specific architecture.

3. Predicted, not measured, activation.. TRIBE v2 produces *predicted* BOLD responses, not actual recordings. While its training objective explicitly aligned predictions with held-out fMRI data, prediction accuracy varies by region and stimulus type, and is bounded above by the irreducible noise of fMRI itself. Our results should be read as "what the best current cortical encoder predicts the brain would do," not as direct neural measurement.

4. Six-region resolution.. The 20,484 cortical vertices were aggregated into six anatomically motivated regions (visual, auditory, language, prefrontal, motor, parietal). This coarsening discards substantial within-region structure — e.g., the fusiform face area within visual cortex, or Broca's vs. Wernicke's areas within language cortex. Vertex-level analyses are possible with the same data and constitute an important extension.

5. Static, decontextualised stimuli.. Real internet consumption is contextual: stimuli arrive in feeds, with surrounding content modulating responses. Our experimental design treats each stimulus as isolated. Sequential or context-conditioned analyses would more closely model real consumption.

6. English-only corpus.. All stimuli were English-language. Cross-linguistic generalisability of the observed effects is unknown, particularly for categories whose linguistic realisation varies substantially across languages (e.g., narrative structure in oral vs. literary cultures).

7. Unequal category sample sizes.. Category sizes range from $N = 88$ (Spatial) to $N = 300$ (Narrative, Factual, Abstract, Social). Smaller categories have wider confidence intervals and lower power in pairwise comparisons. The Spatial category's $\mathrm{ICC}_{\mathrm{proxy}}$ of 0.729 — below the "good" threshold — is consistent with reduced reliability at $N = 88$; findings involving Spatial should

be interpreted cautiously.

8. Category taxonomy not validated.. The 13 categories are author-defined (Section 3.3). Several categories are defined primarily by source corpus rather than by a distinct neurological or psychological construct. Cross-category comparisons should be treated as comparisons of operationally defined groupings, not as comparisons of cognitively validated types.

9. ICC proxy formula is non-standard.. The reproducibility index used in Section 8.2 is an ad-hoc variance-partitioning measure, not the standard ICC(2,1) or ICC(3,1) defined by Shrout & Fleiss (1979). It behaves like a standard ICC in the limit but is not directly comparable to published ICC benchmarks.

10. Single average-brain model; no individual differences.. TRIBE v2 was trained to predict the average neural response of a small number of participants. It does not model individual variability in neural responses. All conclusions apply to a hypothetical "average brain" and cannot speak to individual differences in neural reactivity (e.g., by age, neurotype, or prior media exposure).

9.2 Theoretical Limitations

1. Operationalisation of theoretical predictions.. Our scorecard (Figure 8) reduces complex theoretical predictions to binary or trinary outcomes. More fine-grained operationalisations — e.g., GWT's "ignition" as a temporal phenomenon, IIT's Φ as a graph-theoretic measure — would enable more nuanced tests but require time-series rather than static activation data.

2. Causality.. Our analysis is correlational. Whether high cortical activation in response to Threat-Safety content *causes* downstream behavioural effects (recall, sharing, anxiety) cannot be tested in this design. Combined neural-behavioural studies, ideally with intervention paradigms, are required.

9.3 Future Work

The six extensions originally proposed have made differential progress in this revision; we restate them with current status.

1. **Semantic replication** (*completed — full $N = 30$/CT sweep, ANOVA $p = 0.104$*): The full stratified sweep ($N = 30$/CT, $N = 390$ total) under LLaMA-3.2-3B encoding is reported in Section 8.5. The activation spread is $4.0\times$ wider than hash mode; the ANOVA does not reach significance at this sample size, and the hash-mode and semantic-mode rankings are essentially uncorrelated.

2. **TRIBE v2 robustness checks** (*partially addressed — seq_len robustness + LSA proxy completed; independent cortical encoder pending*): Internal seq_len robustness (Section 8.6) confirms the semantic-mode ordering is not a temporal-integration artefact (Spearman $\rho \geq 0.956$ across seq_len $\in \{4, 8, 16\}$). The LSA text-similarity proxy confirms TRIBE's representational geometry is essentially independent of text surface similarity (Mantel $r = 0.055$, $p = 0.340$). **Remaining**: application of the same corpus to independent cortical encoders (BrainBERT family; Huth et al., 2016 semantic atlas) to determine whether the Social > Abstract > Spatial ordering is a TRIBE v2 property or a brain property.

3. **Vertex-level analysis** (*completed*): Reported in Section 8.1. The six-region aggregation was found to underestimate discriminative power by $\sim 5\times$, and the top-100 most discriminating

vertices distribute across all six anatomical regions.

4. **Temporal dynamics** (*harness ready, sweep deferred*): The sweep harness has been extended to capture TRIBE v2's full 16-timestep output, and the analysis pipeline is in place. We deferred the temporal sweep to the same revision as the full semantic sweep so that both can be run with semantic LLaMA encoding in a single batch — running it now under hash encoding would risk contamination from the same artefact identified in Section 8.5.

5. **Cross-cultural and multilingual replication** (*partially completed*): A small semantic-encoded multilingual sweep is reported in Section 8.4 ($N = 14$ across seven languages); the cross-language ANOVA was preliminarily non-significant. The full 138-stimulus corpus is released with the project (`results/extended/multilingual_corpus.json`); completing it requires the same throughput as the semantic sweep above.

In addition, this revision introduced two new robustness analyses — cross-source ICC (Section 8.2) and the full 13×13 Cohen's d matrix (Section 8.3) — both of which support rather than undermine the hash-mode principal findings; completed the full semantic sweep (Section 8.5), which confirms a wider discriminative signal under proper encoding while showing the specific category rankings shift substantially from the hash-mode baseline; and completed a partial robustness check (Section 8.6) establishing internal seq_len stability and independence from text surface similarity (full cross-encoder replication remains outstanding).

10. Conclusion

We present the first large-scale computational comparison of predicted cortical activation across 13 categories of internet content, using the TRIBE v2 deep fMRI encoder and a corpus of 3,008 stimuli drawn from validated research datasets and live internet sources. A statistically robust effect of content category on predicted whole-brain activation was observed ($F(12, 2995) = 13.51$, $p < 0.0001$, $\eta^2 = 0.051$), with all six cortical regions showing independent significant effects.

Under *hash-mode* encoding, **ThreatSafety content** consistently produced the highest predicted cortical activation and **Narrative content** the lowest. **Multimodal** and **AudioText** content showed the highest relative language-region activation, while **Emotional** content showed the highest relative prefrontal engagement. A dominant cortical gradient (PC1, 96.9% variance) contrasting sensory-language activation against executive-motor activation provides a principled axis for situating content types in neural space.

Update from the full semantic-mode replication and robustness checks.
The complete LLaMA-3.2-3B sweep (Section 8.5) shows a $4.0\times$ wider activation spread under proper semantic encoding, but the rank-ordering of categories is essentially uncorrelated with the hash-mode results ($r = 0.09$, $p = 0.78$). Under semantic encoding, the top three most-activating categories are **Social**, **Abstract**, and **Spatial**; **ThreatSafety** ranks sixth and **Narrative** ninth. The per-category ANOVA does not reach significance at $N = 30/\text{CT}$ ($F(12, 377) = 1.549$, $p = 0.104$) — a larger dataset is needed.
The robustness checks (Section 8.6) confirm that this Social > Abstract > Spatial ordering is stable across temporal integration windows ($\rho \geq 0.956$) and is not explained by text surface similarity (Mantel $r = 0.055$, $p = 0.340$).
We treat the hash-mode results as a baseline establishing *that* category effects exist. The specific identity of the most-activating categories under semantic encoding requires higher-powered replication and validation with independent cortical encoders.

The broader implication is this: **the internet is not a neutral delivery mechanism**. Different content types engage different brain circuits with different intensities. Threatening content is not more viral simply because of algorithms — it may be more viral, in part, because it is more neurologically arresting. Understanding this differential engagement at the level of predictive neural models, rather than self-report surveys, is a precondition for evidence-based digital content policy, platform design, and public health decisions around media consumption.

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